

Phase 4: Model Validation Summary

Comprehensive Validation and Robustness Analysis of Machine Learning Models

Date: March 13, 2026
Project: NOA Prediction System with Machine Learning

Executive Summary

Phase 4 included comprehensive validation of 15 machine learning models through 10 different validation methods. The main goal of this phase is to ensure the reliability, reproducibility, and robustness of the models for clinical use.

Key Findings

Best Model: SVM (Support Vector Machine)

Metric	Result
Best AUC	0.984
95% Confidence Interval	[0.976 - 0.991]
Stability	High (CV: 0.08%)
Noise Robustness	High (-0.02%)

Validation Methods

1. K-Fold Cross-Validation

- 5 and 10 folds
- SVM showed the best performance with AUC 0.985

2. Stratified Analysis

- Coefficient of Variation (CV) computed for all models
- 15/15 models demonstrated high stability

3. LOOCV (50-fold approximation)

- Confirmed K-Fold results
- High agreement between methods

4. Repeated K-Fold (10×10)

- 100 evaluations per model
- Tighter confidence intervals

5. Bootstrap Validation

- 50–100 iterations per model
- 95% confidence intervals computed

6. Nested Cross-Validation

- Separates hyperparameter tuning from evaluation
- Eliminates optimistic bias

7. Overfitting Detection

- 12 models needed regularization tuning
- DecisionTree: severe overfitting (28.35%)

8. Learning Curves

- Bias-variance trade-off analysis
- Identification of optimal models

9. Stability Analysis (10 random seeds)

- All models: CV < 2%
- SVM: most stable (CV: 0.08%)

10. Robustness Test (noise injection)

- Noise levels: 1%, 5%, 10%
- 13/15 models: high robustness

Model Ranking

Rank	Model	AUC	Confidence Interval
1	SVM	0.984	[0.976, 0.991]
2	MLP	0.976	[0.963, 0.985]
3	TabNet	0.975	[0.963, 0.985]
4	CatBoost	0.970	[0.961, 0.980]
5	StackingEnsemble	0.968	[0.956, 0.978]

Key Results

Model Stability

- **All 15 models:** high stability (CV < 2%)
- **SVM:** most stable with CV 0.08%
- **DecisionTree:** most variable with CV 1.25%

Noise Robustness

- **1% noise:** all models robust
- **5% noise:** 14/15 models robust
- **10% noise:** 13/15 models robust

Overfitting

- **No overfitting:** SVM, LogisticRegression, NaiveBayes
 - **Tuning required:** the other 12 models
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Clinical Recommendations

Primary Recommendation

Use SVM as the primary model:

- Best discriminative performance (AUC: 0.984)
- Exceptional stability
- High noise robustness
- No overfitting

Secondary Recommendations

1. **MLP:** for high-noise conditions
 2. **CatBoost:** for gradient-boosting approaches
 3. **TabNet:** when interpretability is needed
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Next Steps: Phase 5

Explainable Artificial Intelligence (XAI)

Phase 5 will include:

1. **Feature Importance Analysis**
 - Identification of key predictive factors
2. **SHAP Analysis**
 - Global- and local-level prediction explanations
3. **Partial Dependence Plots (PDP)**
 - Effect of each variable on the prediction
4. **LIME Explanations**
 - Local model interpretation

5. Clinical Interpretation

- Translation of findings for clinicians

Files Required for the Next Phase

Result Files

- `Phase4_Final_Results.json`
- `kfold_cv_results.csv`
- `bootstrap_validation_results.csv`
- `stability_seeds_analysis.csv`
- `robustness_noise_results.csv`

Model Files

- All `*.joblib` files in `/6_Models/Saved/`

Data Files

- `encoded_dataset.csv`

Conclusion

Phase 4 was completed successfully. The comprehensive validation showed that:

- SVM is the best model, with AUC 0.984
- All models exhibit high stability
- 13 models are robust to noise
- 12 models were tuned to improve generalization

The models are ready to proceed to the Explainable AI phase (Phase 5).

Report prepared: March 2026

NOA ML research team